Classical ILC/DRP Design from an Algebraic (Matrix-Fraction) Perspective – Part 2: Design

Kevin L. Moore

Department of Electrical Engineering
Colorado School of Mines
Golden, Colorado, USA

Pre-Congress Worshop WS-13 – Learning from Data: Principles, Methods, and Emerging Ideas & Applications in Iterative Learning Control

2026 IFAC World Congress
Busan, Republic of Korea
23 August 2026

With Contributions from El-Sharif Omer, Hyo-Sung Ahn, and YangQuan Chen (RIP)

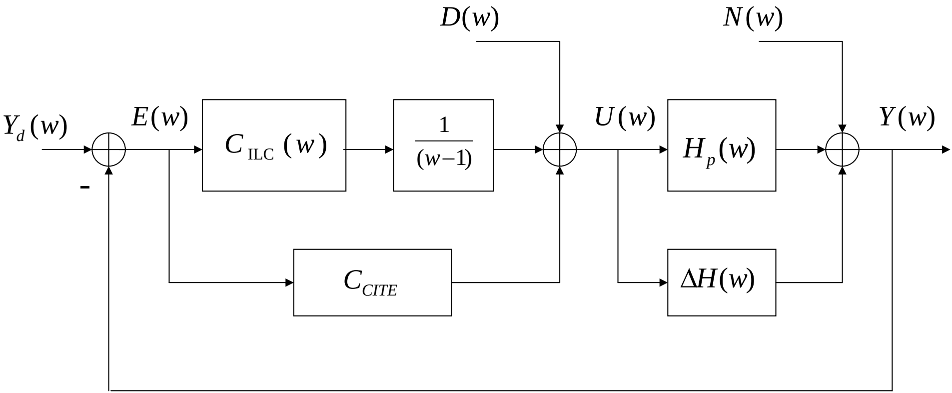
Review from Part 1

- ILC, RC, and DRPs are classes of **multi-pass systems** that can use data to improve their operation from pass-to-pass (ILC is a special case of DRPs)

- In the discrete-time case, **lifted models** provide a useful representation for analysis and design

- By adopting an **operator transform** on the iteration domain (for fixed time), a general **MIMO framework** can be developed for control design problems

Complete Framework



Categorization of Problems

Y_d	D	N	C	H	
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H_p	Classical Arimoto algorithm
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	$H_p(w)$	Rogers'/Owens' multipass problem
$Y_d(z)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	$N(w)$	$C(w)$	$H_p(w)+\Delta(w)$	**Most general problem**
$Y_d(z)$	$D(z)$	0	$C(w)$	$H(w)+\Delta(w)$	Frequency-domain uncertainty
$Y_d(z)$	0	$w(t), v(t)$	$\Gamma(w-1)^{-1}$	H_p	Least quadratic ILC
$Y_d(z)$	0	$w(t), v(t)$	$C(w)$	H_p	Stochastic ILC (general)
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H^I	Interval ILC
$Y_d(z)$	$D(z)$	$w(t), v(t)$	$C(w)$	$H(z)+\Delta H(z)$	Time-domain H_∞ problem
$Y_d(z)$	$D(w)$	$w(k,t), v(k,t)$	$C(w)$	$H(w)+\Delta H(w)$	Iteration-domain H_∞ ILC
$Y_d(z)$	0	$w(k,t), v(k,t)$	$\Gamma(k)$	$H_p+\Delta H(k)$	Iteration-varying uncertainty and control
$Y_d(z)$	0	$\widetilde{H}$	Γ	H_p	Intermittent measurement problem
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

In This Two-Part Talk ...

- Part 0: Brief Overview of two of these paradigms:
  - ILC
  - DRP
- Part 1: Analysis and design framework for such systems
- **Part 2: Design for discrete ILC/DRP systems**
  - **Monotonic convergence in iteration**
    - **Design using current iteration controllers (CITE)**
    - **Design using optimal PD-type algorithms**
    - Iteration-varying gains (not covered today)
  - **Internal model-principle**
  - **Model reference adaptive control (MRAC)**
  - ... and more (not covered today)
    - Robust designs (H_∞, L_1/l_1, L_2/l_2, Interval ILC)
    - ID and Estimation (Kalman filtering in iteration)
    - MPC (norm-optimal - covered in other talks)

Control in 2-D ILC/DRP Systems

We have two axes along which control can be done:

- **Time** (current iteration or CITE control)
  - Can do normal P, PI, PID, or general $C(z)$
  - But "D" is approximation (backwards difference)
- **Iteration** (trial-to-trial control)
  - Again can do P, PI, PID, or general $C(w)$
  - In case of PID
    - "P" is the CITE term
    - "I" is inherent in ILC update (if $Q = I$)
    - True "D" is not possible in iteration (but could also do backwards difference in iteration)

What Information can be Included in the ILC Update?

- Most generally, we can allow:

$$\begin{aligned} u_{k+1}(t) \;=\; f\{ & u_0(t'), u_1(t'), \ldots, u_k(t'), \\ & e_1(t')), e_2(t'), \ldots, e_k(t'), \\ & u_{k+1}(0), u_{k+1}(1), \ldots, u_{k+1}(t-1) \\ & e_{k+1}(1), e_{k+1}(2), \ldots, e_{k+1}(t-1)\} \end{aligned}$$

  where $t' \in [0, N]$.

- That is, in general we can update $u_{k+1}(t)$ using:
  1. Information from all previous trials:
     $\Rightarrow$ Call this "**higher-order in iteration**" if more than one-trial back is used.
  2. Information from the entire time duration of any previous trial:
     $\Rightarrow$ Call this "**higher-order in time**" if filtering is done rather than using a single time instance.
     $\Rightarrow$ Note this allows non-causal signal processing – a key reason ILC works.
  3. Information up to time $t-1$ on the current trial:
     $\Rightarrow$ Call this "**current cycle feedback**."

Higher-Order vs. First-Order

- Is there any reason to use higher-order ILC algorithms (in time or in iteration)?
- Maybe? Because of convergence? No, consider:

1. Classical Arimoto D-type ILC (for relative degree 1):

$$u_{k+1}(t) = u_k(t) + \gamma \frac{\mathrm{d}}{\mathrm{d}t} e_k(t)$$

2. PID-type ILC:

$$u_{k+1}(t) = u_k(t) + k_P e_k(t) + k_I \int_0^t e_k(\tau)\mathrm{d}\tau + \gamma \frac{\mathrm{d}}{\mathrm{d}t} e_k(t)$$

Higher-Order vs. First-Order (cont.)

- For both, the convergence condition is that

$$|1 - \gamma h_1| < 1$$

where h_1 is the first Markov (non-zero) parameter. This ensures $e_k(t) \to 0$ as $k \to \infty$ $\forall t$.

- Note: does not involve k_P, k_I, or with the system matrix A, either!

- That is, first-order in time and iteration is adequate to realize convergence.

- Something must be missing ...

- The answer is: "how the convergence is achieved."

Common Control Structures

In the remainder of this section we will consider several iteration-domain controllers:

- Two P-type "Arimoto" ILC schemes defined in the literature
  - $u_{k+1}(t) = u_k(t) + k_p e_k(t+1)$
  - $u_{k+1}(t) = u_k(t) + k_p e_k(t)$
- Variations on PD-type ILC schemes:
  - $u_{k+1}(t) = u_k(t) + k_p e_k(t) + k_d(e_k(t+1) - e_k(t))$
  - $u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta e_k(t))$
- These PD-type schemes effectively produce a true "D" term along the time axis in the limit as $k \to \infty$

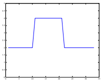
Two Examples

- Consider two systems, each stable, minimum phase, with the same ILC update law

$$u_{k+1}(t) = u_k(t) + 0.9e_k(t+1)$$

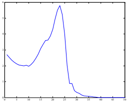
1. $y_k(t+1) = -.2y_k(t) + .0125y_k(t-1) + u_k(t) - 0.9u_k(t-1)$
2. $y_k(t+1) = -.2y_k(t) + .0125y_k(t-1) + u_k(t) + 0.1u_k(t-1)$

- Each is asked to track the following signal:

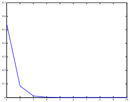


- The convergence condition guarantees that both converge, but ...

System 1 does not converge monotonically (in 2-norm):



System 2 does converge monotonically (in 2-norm):



Question: Why do the two systems learn differently?

Comments

- In the literature it has been shown that ILC achieves monotonic convergence for the λ-norm (time-weighted-norm) of the tracking error.
- However, in general the ∞-norm and 2-norm will often increase to a huge value before converging.
- Such ILC transients are typically not acceptable!
- It is not enough to ensure that $e_k(t) \rightarrow 0$ as $k \rightarrow \infty$. Rather, we would like the convergence to be monotonic.
- And, the norm topology should be physically meaningful.

Designing for Monotonic Convergence

- Using "higher-order-in-time" algorithms, that is, proper design of the ILC update filters or algorithms, can give monotonic convergence through:
  1. Higher-order-in-time (causal) current-cycle feedback
  2. Non-causal filtering of the error from the previous trial
  3. Time-varying ILC gains (not covered today)

Designing for Monotonic Convergence

- Using "higher-order-in-time" algorithms, that is, proper design of the ILC update filters or algorithms, can give monotonic convergence through:
  1. Higher-order-in-time (causal) current-cycle feedback
  2. Non-causal filtering of the error from the previous trial
  3. Time-varying ILC gains (not covered today)
- Additionally, a frequency domain perspective can be used to add a "Q-filter" such as

$$U_{k+1} = Q(U_k + LE_k)$$

can result in monotonic convergence, but at a loss of steady-state tracking

Designing for Monotonic Convergence

- Using "higher-order-in-time" algorithms, that is, proper design of the ILC update filters or algorithms, can give monotonic convergence through:
  1. Higher-order-in-time (causal) current-cycle feedback
  2. Non-causal filtering of the error from the previous trial
  3. Time-varying ILC gains (not covered today)
- Additionally, a frequency domain perspective can be used to add a "Q-filter" such as

$$U_{k+1} = Q(U_k + LE_k)$$

  can result in monotonic convergence, but at a loss of steady-state tracking
- Finally, Norm-Optimal ILC (NOILC) will generally converge monotonically

Monotonic Convergence Condition

- For the Arimoto-update ILC algorithm, the ILC scheme converges (monotonically) if the induced operator norm satisfies:

$$\|I - \gamma H_p\|_i < 1.$$

- Likewise, a NAS for convergence is:

$$|1 - \gamma h_1| < 1.$$

- Combining these, we can show that for a given gain γ, convergence implies monotonic convergence in the ∞-norm if

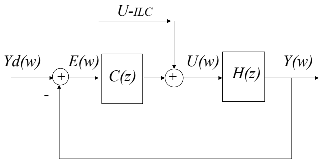
$$|h_1| > \sum_{j=2}^{N} |h_j|.$$

- Note this condition is independent of γ, but instead puts restrictions on the plant.

Control plant to meet convergence condition

Method 1: Current-Cycle Feedback

- Case A:

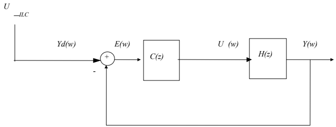


- Plant seen by the ILC algorithm:

$$H_{cl}^{A} = \frac{H(z)}{1 + C(z)H(z)}.$$

Method 1 (cont.)

- Case B:



- Plant seen by the ILC algorithm:

$$H_{cl}^{B} = \frac{C(z)H(z)}{1 + C(z)H(z)}.$$

Idea: pick $C(z)$ so closed-loop is FIR and monotonic convergence condition is met

FIR Approach

- Let

$$\begin{aligned} H(z) &= h_1 z^{-1} + h_2 z^{-2} + \cdots, \\ C(z) &= c_0 + c_1 z^{-1} + c_2 z^{-2} + \cdots, \end{aligned}$$

- For Case A the monotonic convergence condition can be shown to be:

$$|h_1| > \sum_{i=2}^{N} |h_i - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j}|.$$

- For Case B the monotonic convergence condition can be shown to be:

$$|c_0 h_1| > \sum_{i=2}^{N} |\sum_{j=1}^{i} h_j c_{i-j} - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j}|.$$

FIR Approach (cont.)

- For both Case A and Case B a controller always exists to give a closed-loop system that satisfies the monotone convergence condition.

- For example, for Case A we can pick:

$$|h_i - h_1 \sum_{j=1}^{i-1} h_j c_{i-1-j}| = 0,$$

- That is, we solve recursively the following:

$$\begin{aligned} 0 &= |h_2 - h_1 c_0|, \\ 0 &= |h_3 - h_1(h_1 c_1 + h_2 c_0)|, \\ 0 &= |h_4 - h_1(h_1 c_2 + h_2 c_1 + h_3 c_0)|, \\ &\vdots \end{aligned}$$

- Then the system will have monotonic ILC convergence whenever ILC converges.

Comments

- With classical Arimoto-type ILC algorithms, the equivalence of ILC convergence with monotonic ILC convergence depends on the characteristics of the plant
- If a plant does not have the characteristics that ensure such monotonic convergence it is possible to "condition" the plant prior to the application of ILC using current cycle-feedback

Comments

- With classical Arimoto-type ILC algorithms, the equivalence of ILC convergence with monotonic ILC convergence depends on the characteristics of the plant
- If a plant does not have the characteristics that ensure such monotonic convergence it is possible to “condition” the plant prior to the application of ILC using current cycle-feedback
- One such current-cycle feedback strategy was presented: an FIR design that results in high-order controller, with monotonic convergence always guaranteed, but possible robustness problems
- An IIR design approach can be developed with better robustness, but a solution is not always guaranteed

Design for Monotone Convergence

Using the monotonic convergence condition, we have derived ILC algorithm designs using higher-order time-domain filtering to achieve monotonic convergence three ways:

1. Causal current-cycle feedback
2. **Non-causal filtering of the error from the previous trial (optimal design of L for PD-type ILC)**
   - **Examples**
   - **Optimal PD-type ILC Scheme: How to Design**
   - **Optimal PD-type ILC Scheme: Averaged Derivative**
   - **Remarks**
3. Time-varying ILC gain.

Examples: PD-Type ILC

Simulation scenarios:

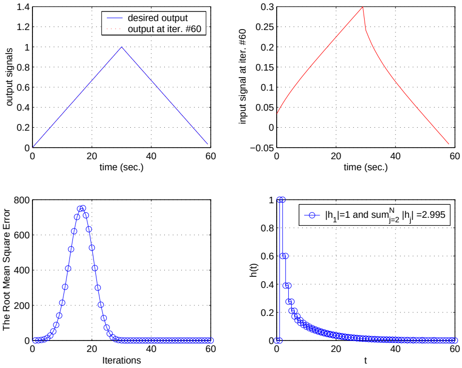
- Second order IIR models are used. All initial conditions are set to 0.
- All plants have $h_1 = 1$, so we fix γ=0.9 such that $|1 - \gamma h_1| < 1$.
- We fix N=60 and max number of iterations = 60.
- The desired trajectory is a triangle given by

$$y_d(t) = \begin{cases} 2t/N & , \; i = 1, \cdots, N/2 \\ 2(N-t)/N & , \; i = N/2+1, \cdots, N. \end{cases}$$

- We compare $u_{k+1}(t) = u_k(t) + \gamma e_k(t+1)$ with $u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t))$

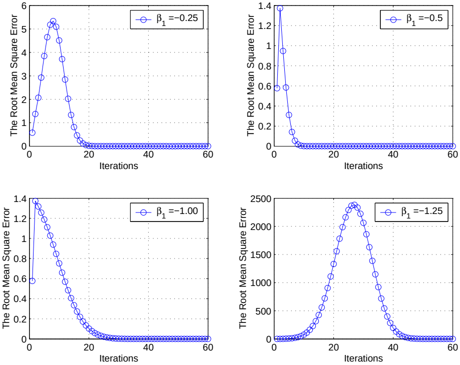
Plant 1a. Stable lightly damped. $H_1(z) = \frac{z-0.8}{(z-0.5)(z-0.9)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1),\ \gamma = 0.9$$



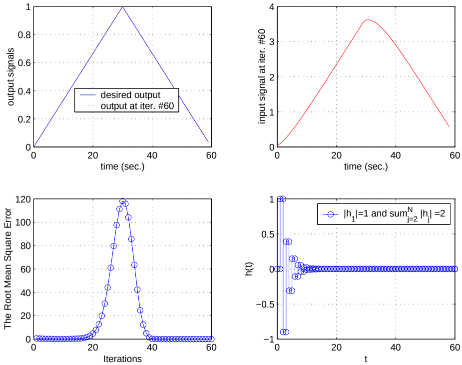
Plant 1b. Stable lightly damped. $H_1(z) = \frac{z-0.8}{(z-0.5)(z-0.9)}$.

$$u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t)) \text{ with } \gamma = 0.9 \text{ fixed and } \beta_1 \text{ shown on the plots.}$$



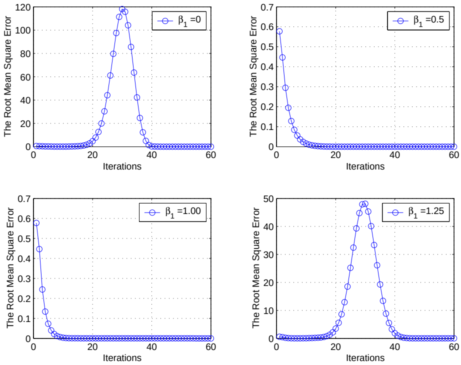
Plant 2a. Stable oscillatory. $H_2(z) = \frac{z-0.8}{(z-0.5)(z+0.6)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1),\ \gamma = 0.9$$



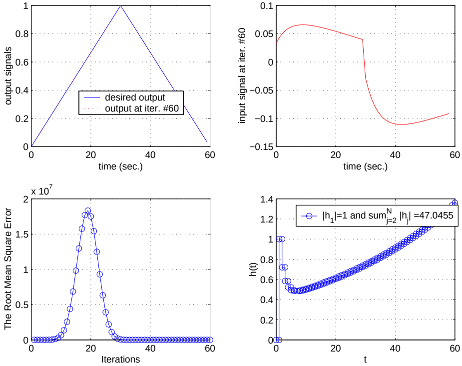
Plant 2b. Stable oscillatory. $H_2(z) = \frac{z-0.8}{(z-0.5)(z+0.6)}$.

$$u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t)) \text{ with } \gamma = 0.9 \text{ fixed and } \beta_1 \text{ shown on the plots.}$$



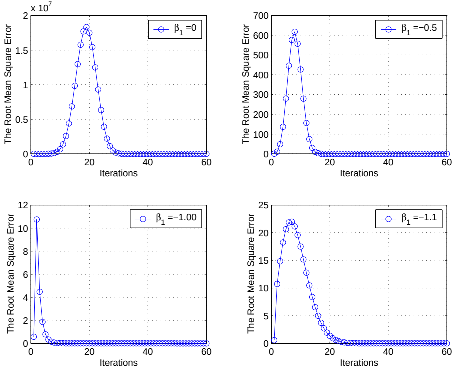
Plant 3a. Slightly unstable. $H_3(z) = \frac{z-0.8}{(z-0.5)(z-1.02)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1),\ \gamma = 0.9$$



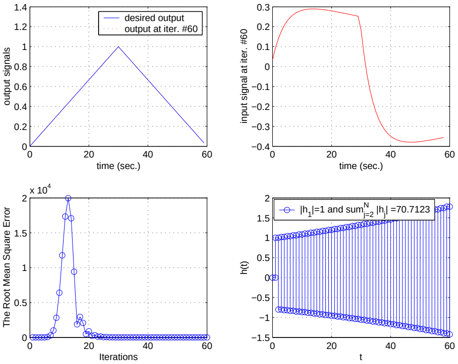
Plant 3b. Slightly unstable. $H_3(z) = \frac{z-0.8}{(z-0.5)(z-1.02)}$.

$$u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t)) \text{ with } \gamma = 0.9 \text{ fixed and } \beta_1 \text{ shown on the plots.}$$



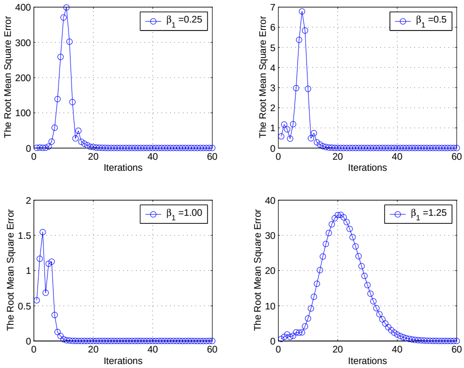
Plant 4a. Unstable oscillating. $H_4(z) = \frac{z-0.8}{(z+1.01)(z-1.01)}$.

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1),\ \gamma = 0.9$$



Plant 4b. Unstable oscillating. $H_4(z) = \frac{z-0.8}{(z+1.01)(z-1.01)}$.

$$u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t)) \text{ with } \gamma = 0.9 \text{ fixed and } \beta_1 \text{ shown on the plots.}$$



Examples: PD-Type ILC (cont.)

- For $u_{k+1}(t) = u_k(t) + \gamma(e_k(t+1) - \beta_1 e_k(t))$ we conclude that:
  - Monotone convergence is possible for the right values of γ and β.
  - Can relate "overshoot" in convergence for some values of β to zeros in the iteration domain.
- In fact, further, can show:
  - Better convergence behavior is possible with $\beta < 0$.
  - How to pick the optimal β.
- In these simulations we used a simple structure. More generally, we can show how to pick a general lower triangular Toeplitz L (i.e, design of $L(z)$) to find the optimal ILC filter for monotonic convergence.

Comments about Monotonic ILC

- Guaranteeing monotonic convergence of an ILC system is practically important and is theoretically desirable.

- Both higher-order-in-time and first-order-in-iteration have been analyzed with respect to monotonic convergence.

- We found also found that time-varying learning gains could be used for monotonic convergence. This is practically important because without using causal and noncausal bands, the monotonic convergence can be achieved.

- If we just consider the time domain, it is very difficult to guarantee the monotonic condition, while in the iteration domain, the monotonic condition can be achieved relatively easily.

- Various monotonic convergence conditions under various ILC algorithms have been studied. In particular, we have shown that the LMI tool box can be used to design monotonically-convergent ILC algorithms.

For Further Reading on Optimal PD Design

- "An Observation about Monotonic Convergence in Discrete-Time, P-Type Iterative Learning Control," Kevin L. Moore, in *Proceedings of 2001 IEEE International Symposium on Intelligent Control*, Mexico City, Mexico, September 2001.
- "PI-type Iterative Learning Control Revisited," YangQuan Chen and Kevin L. Moore in *Proceedings of the 2002 American Control Conference*, Anchorage, Alaska, May 8-10, 2002.
- "An Investigation on the Monotonic Convergence of an High-Order Iterative Learning Update Law," Kevin L. Moore and YangQuan Chen, in *Proceedings of the 2002 15th IFAC World Congress*, Barcelona, Spain, July 21-26, 2002.
- "Harnessing the Repetitiveness in Iterative Learning Control," YangQuan Chen and Kevin L. Moore, in *Proceedings of 2002 IEEE Conference on Decision and Control*, Las Vegas, NV, Dec 22-14, 2002.
- "An Optimal Design of PD-type Iterative Learning Controller With Monotonic Convergence," YangQuan Chen and Kevin L. Moore, in *Proceedings of the 17th IEEE International Symposium on Intelligent Control, IEEE ISIC'02*, Vancouver, British Columbia October 27-30, 2002.
- "Feedback Controller Design to Ensure Monotonic Convergence in Discrete-Time, P-Type Iterative Learning Control," Kevin L. Moore, YangQuan Chen, and Vikas Bahl in *Proceedings of 4th Asian Control Conference*, Singapore, Sept 2002.
- "A Separative High-Order Framework for Monotonic Convergent Iterative Learning Controller Design," Kevin L. Moore and YangQuan Chen, in *Proceedings of the 2003 American Control Conference*, Denver, Colorado, June 4 to 6, 2003.
- "Monotonically Convergent Iterative Learning Control for Linear Discrete-Time Systems," Kevin L. Moore, YangQuan Chen, and Vikas Bahl, *Automatica,* vol. 41, pp. 1529-1537 2005.
- "LMI Approach to Iterative Learning Control Design," Hyo-Sung Ahn, K.L. Moore, and Y.Q. Chen, *Proceedings of 2006 IEEE Mountain Workshop on Adaptive and Learning Systems*, Utah State University, Logan, Utah, July 2006.

Internal Model Principle -1

- Suppose that the reference signal $y_k^d(t)$ has the lifted representation

$$Y_k^d = (y_k(0), y_k(1), \cdots, y_k(N-1))^T$$

with a corresponding complex iteration-domain representation

$$Y^d(w) = (Y_0^d(w), Y_1^d(w), \cdots, Y_N^d(w))^T$$

Internal Model Principle -1

- Suppose that the reference signal $y_k^d(t)$ has the lifted representation

$$Y_k^d = (y_k(0), y_k(1), \cdots, y_k(N-1))^T$$

with a corresponding complex iteration-domain representation

$$Y^d(w) = (Y_0^d(w), Y_1^d(w), \cdots, Y_N^d(w))^T$$

- Let $\phi(w)$, a scalar, be the least common denominator of the unstable poles of each $Y_i^d(w)$

Internal Model Principle -2

- Let the denominator of the controller be given as

$$D_c(w) = \phi(w)\bar{D}_c(w)$$

Internal Model Principle -2

- Let the denominator of the controller be given as

$$D_c(w) = \phi(w)\bar{D}_c(w)$$

- Applying the final value theorem we can conclude that the steady-state error

$$E_{ss} = \lim_{k\to\infty} E_{k+1} \;=\; \lim_{w\to 1}(w-1)E(w) = 0$$

if the zeros of the closed-loop denominator matrix $D_{cl(w)}$ are stable, where

$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

Internal Model Principle -2

- Let the denominator of the controller be given as

$$D_c(w) = \phi(w)\bar{D}_c(w)$$

- Applying the final value theorem we can conclude that the steady-state error

$$E_{ss} = \lim_{k\to\infty} E_{k+1} \quad = \quad \lim_{w\to 1}(w-1)E(w) = 0$$

if the zeros of the closed-loop denominator matrix $D_{cl(w)}$ are stable, where

$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

- We call this the *Internal Model Principle for ILC/DRPs*

Simultaneous Stability and Tracking -1

- To track an iteration-varying reference signal we need the zeros of

$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

to be stable

Simultaneous Stability and Tracking -1

- To track an iteration-varying reference signal we need the zeros of

$$D_{cl}(w) = \phi(w)\bar{D}_c(w)(wI - H^y) + N_c(w)wH^u$$

to be stable

- Note that the plant

$$P(w) = wH^u(wI - H^y)^{-1}$$

is coprime in the space of w-transforms

Simultaneous Stability and Tracking -2

- It follows from standard results that there always exists a stabilizing tracking controller given by

$$C(w) = \frac{1}{\phi(w)} \bar{D}_c^{-1}(w) N_c(w)$$

where $\bar{D}_c(w)$ and $N_c(w)$ can be chosen to make the system stable by solving a Bezout equation

Simultaneous Stability and Tracking -2

- It follows from standard results that there always exists a stabilizing tracking controller given by

$$C(w) = \frac{1}{\phi(w)} \bar{D}_c^{-1}(w) N_c(w)$$

where $\bar{D}_c(w)$ and $N_c(w)$ can be chosen to make the system stable by solving a Bezout equation

- We further point out that ILC/DRP control algorithms that do not honor this internal model principle are not robust with respect to tracking performance

Example 1: Constant Reference Signals -1

- For a constant reference we have $\phi(w) = (w - 1)$

Example 1: Constant Reference Signals -1

- For a constant reference we have $\phi(w) = (w-1)$
- Then minimal-order stabilizing, tracking compensator can be obtained with the structure

$$C(w) = \frac{1}{(w-1)}(N_1 w + N_0)$$

Example 1: Constant Reference Signals -1

- For a constant reference we have $\phi(w) = (w-1)$
- Then minimal-order stabilizing, tracking compensator can be obtained with the structure

$$C(w) = \frac{1}{(w-1)}(N_1 w + N_0)$$

- Closed-loop system stability is defined by the zeros of

$$\begin{aligned} D_{cl}(w) &= (w-1)(wI - H^y) + (N_1 w + N_0)wH^u \\ &= (I + N_1 H^u)w^2 + (N_0 H^u - I - H^y)w + H^y \end{aligned}$$

Example 1: Constant Reference Signals -2

- In the special case where the control is an Arimoto-style update equation

$$u_{k+1}(t) = u_k(t) + \gamma_1 e_{k+1}(t) + \gamma_0 e_k(t)$$

the gains N_1 and N_0 are diagonal matrices and stability is given by the roots of

$$(1 + \gamma_1 h_0^u)w^2 + (\gamma_0 h_0^u - 1 - h_0^y)w + h_0^y = 0$$

Comments

- The resulting controller structure has relative degree zero in the w-domain
- Thus, the controller implicitly involves both past pass information (memory) as well as current pass control

Comments

- The resulting controller structure has relative degree zero in the w-domain
- Thus, the controller implicitly involves both past pass information (memory) as well as current pass control
- The claim of arbitrary pole placement is true for the controller shown, except we cannot place closed-loop poles at $w = 0$
  - It is possible to overcome this limitation by making the controller order larger

Simulation -1

- Consider the plant

$$Y_{k+1}(z) = H^u(z)U_{k+1}(z) + H^y(z)Y_k(z)$$

where

$$\begin{aligned} H^u(z) &= \frac{1.2 + 1.24z^{-1} + 0.085z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \\ H^y(z) &= \frac{0.8 + 0.66z^{-1} + 0.09z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \end{aligned}$$

Simulation -1

- Consider the plant

$$Y_{k+1}(z) = H^u(z)U_{k+1}(z) + H^y(z)Y_k(z)$$

where

$$\begin{aligned} H^u(z) &= \frac{1.2 + 1.24z^{-1} + 0.085z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \\ H^y(z) &= \frac{0.8 + 0.66z^{-1} + 0.09z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \end{aligned}$$

- Let $y^d(t) = 3 + \sin(0.1t) + 1.5\sin(0.5(t-1))$ on the time interval $[1, 30]$

Simulation -2

- Consider two update (or control) laws:

    1. An internal model controller of the Arimoto form described above

$$u_{k+1}(t) = u_k(t) + 1.5e_k(t) - 0.667e_{k-1}(t)$$

Simulation -2

- Consider two update (or control) laws:

1. An internal model controller of the Arimoto form described above

$$u_{k+1}(t) = u_k(t) + 1.5e_k(t) - 0.667e_{k-1}(t)$$

2. A unit memory feedback designed for deadbeat convergence

$$u_{k+1}(t) = -\frac{H^y(z)}{H^u(z)}y_k(t) + \frac{1}{H^u(z)}y^d(t)$$

Simulation -3

- To study robustness, after seventy-five iterations the plant is changed to

$$
\begin{aligned}
H^u(z) &= \frac{1.2632 + 1.3053z^{-1} + 0.0895z^{-2}}{1 + 0.2105z^{-1} - 0.0132z^{-2}} \\
H^y(z) &= \frac{0.8421 + 0.6947z^{-1} + 0.0947z^{-2}}{1 + 0.2105z^{-1} - 0.0132z^{-2}}
\end{aligned}
$$

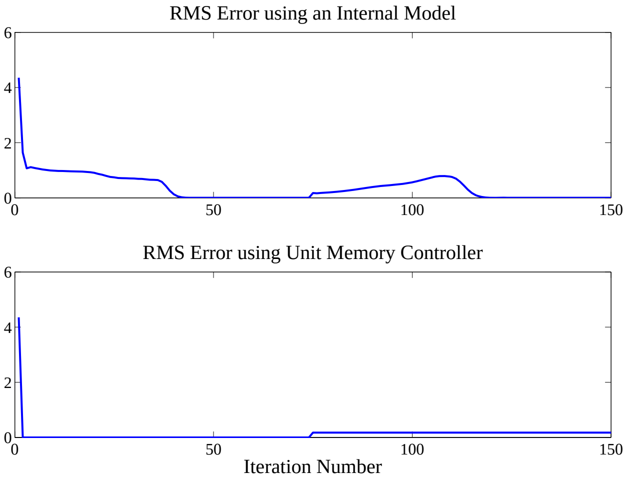
Simulation -3

- To study robustness, after seventy-five iterations the plant is changed to

$$
\begin{aligned}
H^u(z) &= \frac{1.2632 + 1.3053z^{-1} + 0.0895z^{-2}}{1 + 0.2105z^{-1} - 0.0132z^{-2}} \\
H^y(z) &= \frac{0.8421 + 0.6947z^{-1} + 0.0947z^{-2}}{1 + 0.2105z^{-1} - 0.0132z^{-2}}
\end{aligned}
$$

- Both control laws converge for the nominal plant, but only the internal model-based controller responds properly to the perturbation in the plant

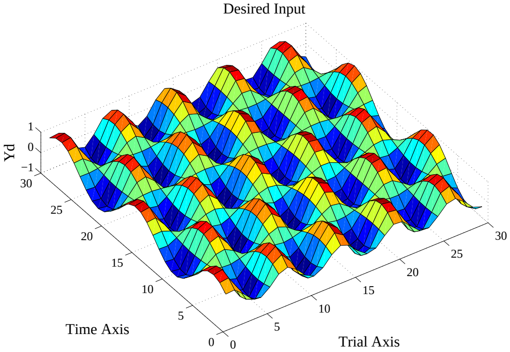
Simulation -4



Example: Sinusoidal Reference -1

- Consider the case when we want to track a signal that varies sinusoidally in iteration

$$y_k^d(t) = \sin(k)\sin(0.4t) \Leftrightarrow Y_t^d(w) = \sin(0.4t)\frac{0.84w}{w^2 - 1.08w + 1}$$



Example: Sinusoidal Reference -2

- Use the update law

$$u_{k+1}(t) = 1.0806u_k(t) - u_{k-1}(t) + e_k(t) - 1.167e_{k-1}(t) + 0.667e_{k-2}(t)$$

which corresponds to the iteration w-domain controller

$$\bar{C}(w) = \frac{1}{w^3 - 1.0806w^2 + w}(N_2 w^2 + N_1 w + N_0)$$

where $N_2 = I$, $N_1 = diag(-1.167)$, and $N_0 = diag(0.667)$

Example: Sinusoidal Reference -2

- Use the update law

$$u_{k+1}(t) = 1.0806u_k(t) - u_{k-1}(t) + e_k(t) - 1.167e_{k-1}(t) + 0.667e_{k-2}(t)$$

  which corresponds to the iteration w-domain controller

$$\bar{C}(w) = \frac{1}{w^3 - 1.0806w^2 + w}(N_2 w^2 + N_1 w + N_0)$$

  where $N_2 = I$, $N_1 = diag(-1.167)$, and $N_0 = diag(0.667)$
- These gains give closed-loop poles $w = 0, 0.3403 \pm j0.5909$, chosen to give a quick settling time and minimum overshoot

Example: Sinusoidal Reference -2

- Use the update law

$$u_{k+1}(t) = 1.0806u_k(t) - u_{k-1}(t) + e_k(t) - 1.167e_{k-1}(t) + 0.667e_{k-2}(t)$$

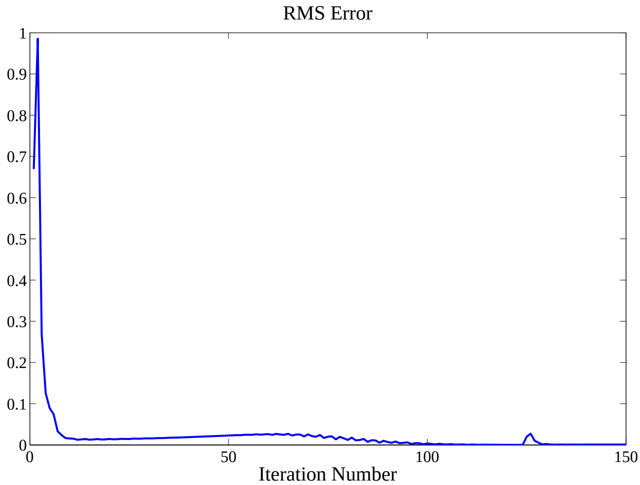
which corresponds to the iteration w-domain controller

$$\bar{C}(w) = \frac{1}{w^3 - 1.0806w^2 + w}(N_2 w^2 + N_1 w + N_0)$$

where $N_2 = I$, $N_1 = diag(-1.167)$, and $N_0 = diag(0.667)$

- These gains give closed-loop poles $w = 0, 0.3403 \pm j0.5909$, chosen to give a quick settling time and minimum overshoot
- Again, the IMP controller gives good robustness to plant perturbations at $k = 125$

Example: Sinusoidal Reference -3



Model Reference Adaptive Control for DRP

- Let the DRP be $Y_p(k+1) = H^u U(k+1) + H^y Y_p(k)$ with corresponding transfer matrix $G_p(w) = w(wI - H^y)^{-1} H^u$

Model Reference Adaptive Control for DRP

- Let the DRP be $Y_p(k+1) = H^u U(k+1) + H^y Y_p(k)$ with corresponding transfer matrix $G_p(w) = w(wI - H^y)^{-1} H^u$
- Let the reference model describing the desired behavior of the DRP be $Y_m(k+1) = B_m R(k) + A_m Y_m(k)$ with corresponding transfer matrix of the reference model $G_m(w) = (wI - A_m)^{-1} B_m$

Model Reference Adaptive Control for DRP

- Let the DRP be $Y_p(k+1) = H^u U(k+1) + H^y Y_p(k)$ with corresponding transfer matrix $G_p(w) = w(wI - H^y)^{-1} H^u$
- Let the reference model describing the desired behavior of the DRP be $Y_m(k+1) = B_m R(k) + A_m Y_m(k)$ with corresponding transfer matrix of the reference model $G_m(w) = (wI - A_m)^{-1} B_m$
  (note that we use $R(k)$ rather than $R(k+1)$ with no loss of generality)

Model Reference Adaptive Control for DRP

- Let the DRP be $Y_p(k+1) = H^u U(k+1) + H^y Y_p(k)$ with corresponding transfer matrix $G_p(w) = w(wI - H^y)^{-1} H^u$
- Let the reference model describing the desired behavior of the DRP be $Y_m(k+1) = B_m R(k) + A_m Y_m(k)$ with corresponding transfer matrix of the reference model $G_m(w) = (wI - A_m)^{-1} B_m$
  (note that we use $R(k)$ rather than $R(k+1)$ with no loss of generality)
- Our goal is a control law that makes $E(k) = Y_p(k) - Y_m(k)$ go to zero

Model Reference Adaptive Control for DRP

- Let the DRP be $Y_p(k+1) = H^u U(k+1) + H^y Y_p(k)$ with corresponding transfer matrix $G_p(w) = w(wI - H^y)^{-1} H^u$
- Let the reference model describing the desired behavior of the DRP be $Y_m(k+1) = B_m R(k) + A_m Y_m(k)$ with corresponding transfer matrix of the reference model $G_m(w) = (wI - A_m)^{-1} B_m$
  (note that we use $R(k)$ rather than $R(k+1)$ with no loss of generality)
- Our goal is a control law that makes $E(k) = Y_p(k) - Y_m(k)$ go to zero
- We assume only the sign of h_0^u

MRAC Control Law

- We propose the following control law

$$U(k+1) = -KY_p(k) + LR(k)$$

MRAC Control Law

- We propose the following control law

$$U(k+1) = -KY_p(k) + LR(k)$$

- With this controller the closed-loop system is

$$G_{cl}(w) = [wI - (H^y - H^u K)]^{-1} H^u L$$

MRAC Control Law

- We propose the following control law

$$U(k+1) = -KY_p(k) + LR(k)$$

- With this controller the closed-loop system is

$$G_{cl}(w) = [wI - (H^y - H^u K)]^{-1} H^u L$$

- It can be seen that we can make $G_{cl} = G_m$ if there exist K^* and L^* to satisfy the matching conditions

$$H^y - H^u K^* = A_m, \qquad H^u L^* = B_m$$

MRAC Control Law

- We propose the following control law

$$U(k+1) = -KY_p(k) + LR(k)$$

- With this controller the closed-loop system is

$$G_{cl}(w) = [wI - (H^y - H^u K)]^{-1} H^u L$$

- It can be seen that we can make $G_{cl} = G_m$ if there exist K^* and L^* to satisfy the matching conditions

$$H^y - H^u K^* = A_m, \qquad H^u L^* = B_m$$

- In DRP a solution always exists due to the Toeplitz nature of H^u and H^y

The Adaptive Update Rule

The gains K and L are updated according to

$$\begin{bmatrix} K(k+1) \\ L(k+1) \end{bmatrix} = \begin{bmatrix} K(k) \\ L(k) \end{bmatrix} +$$

$$\begin{bmatrix} B_m^T PE(k) sign(l) & 0 \\ 0 & B_m^T PE(k) sign(l) \end{bmatrix} \Omega(k)$$

The Adaptive Update Rule

The gains K and L are updated according to

$$\begin{bmatrix} K(k+1) \\ L(k+1) \end{bmatrix} = \begin{bmatrix} K(k) \\ L(k) \end{bmatrix} +$$

$$\begin{bmatrix} B_m^T PE(k) sign(l) & 0 \\ 0 & B_m^T PE(k) sign(l) \end{bmatrix} \Omega(k)$$

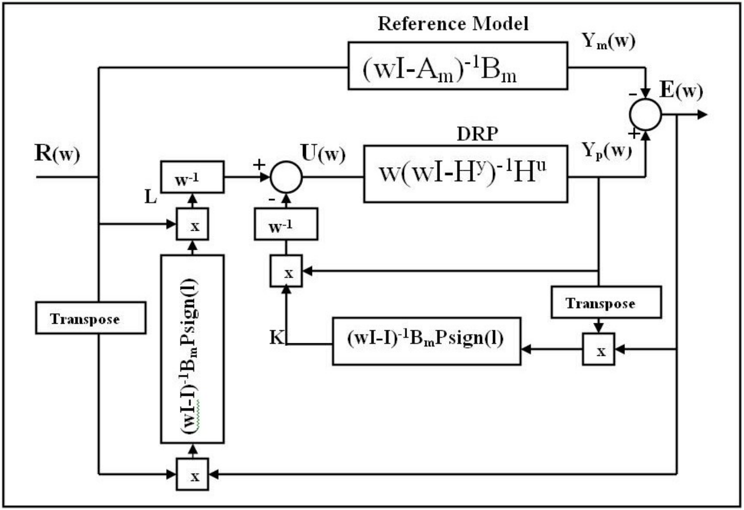
where

$$\Omega(k) = \begin{bmatrix} Y_p^T(k) & -R^T(k) \end{bmatrix}^T$$

$$sign(l) = \begin{cases} 1 & \text{if } L^* \text{ is positive definite} \\ -1 & \text{if } L^* \text{ is negative definite} \end{cases}$$

and $P = P^T > 0$ satisfies $A_m^T P A_m - P = -QQ^T$ for some $Q = Q^T > 0$

Generic MRAC Structure Applied to DRP



MRAC Convergence

- Lyapunov analysis can be used to show
  - The error between the output of the reference model and the output of the plant goes to zero

MRAC Convergence

- Lyapunov analysis can be used to show
  - The error between the output of the reference model and the output of the plant goes to zero
  - If the signal vector $\Omega(k) = [Y_p^T(k) \ \ - R^T(k)]^T$ is persistently exciting then the gains also converge

MRAC Examples

- Consider the same plant as above:

$$Y_p(k+1,z) = H^u(z)U(k+1,z) + H^y(z)Y(k,z)$$

where

$$\begin{aligned} H^u(z) &= \frac{1.2 + 1.24z^{-1} + 0.085z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \\ H^y(z) &= \frac{0.8 + 0.66z^{-1} + 0.09z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \end{aligned}$$

MRAC Examples

- Consider the same plant as above:

$$Y_p(k+1,z) = H^u(z)U(k+1,z) + H^y(z)Y(k,z)$$

where

$$\begin{aligned} H^u(z) &= \frac{1.2 + 1.24z^{-1} + 0.085z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \\ H^y(z) &= \frac{0.8 + 0.66z^{-1} + 0.09z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \end{aligned}$$

- We use the pass length interval $t \in [1, 30]$

MRAC Examples

- Consider the same plant as above:

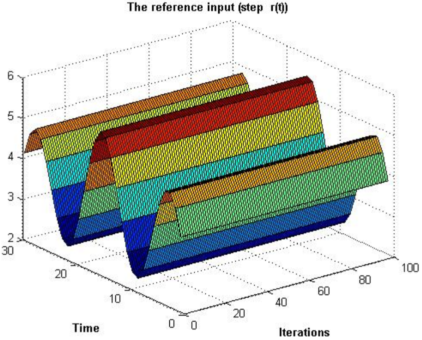
$$Y_p(k+1,z) = H^u(z)U(k+1,z) + H^y(z)Y(k,z)$$

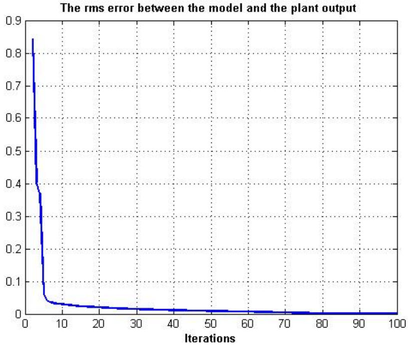
where

$$\begin{aligned} H^u(z) &= \frac{1.2 + 1.24z^{-1} + 0.085z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \\ H^y(z) &= \frac{0.8 + 0.66z^{-1} + 0.09z^{-2}}{1 + 0.2z^{-1} - 0.0125z^{-2}} \end{aligned}$$

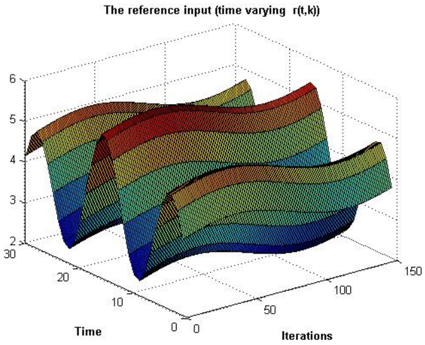
- We use the pass length interval $t \in [1, 30]$
- Reference model uses $A_m = -0.2I_{30\times 30}$ and $B_m = 2.0I_{30\times 30}$

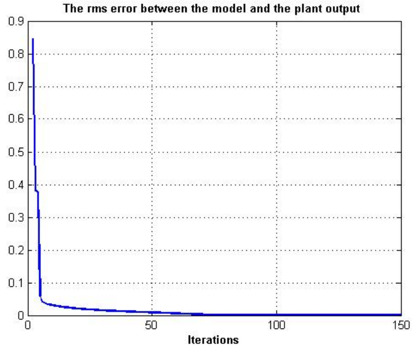
Case 1: $r(t) = 3 + \sin(0.1t) + 1.5\sin(0.5t)$





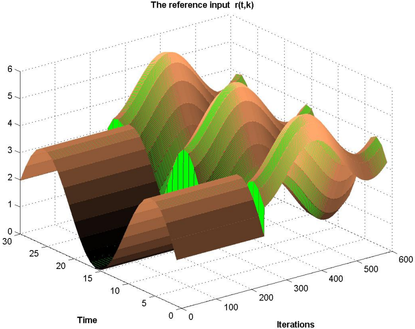
Case 2: $r(t,k) = 3 + \sin(0.1t) + 1.5\sin(0.5t) + 0.3\sin(0.04k)$

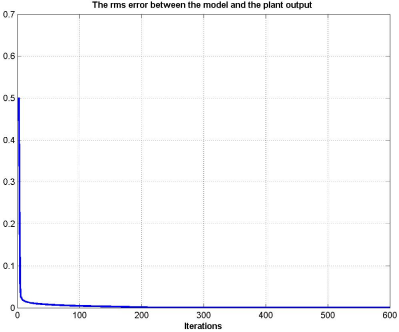




Case 3: Reference Change After Learning

- Reference is changed after 300 iterations





Algebraic (Matrix Fraction) Approach to ILC/DRP

- Part 0: Brief Overview of two of these paradigms:
  - ILC
  - DRP
- Part 1: Analysis and design framework for such systems
- Part 2: Design for discrete ILC/DRP systems
  - Monotonic convergence in iteration
    - Design using current iteration controllers (CITE)
    - Design using optimal PD-type algorithms
    - Iteration-varying gains (not covered today)
  - Internal model-principle
  - Model reference adaptive control (MRAC)
  - ... and more (not covered today)
    - Robust designs (H_∞, L_1/l_1, L_2/l_2, Interval ILC)
    - ID and Estimation (Kalman filtering in iteration)
    - MPC (norm-optimal - covered in other talks)

Future Work

- Future work can consider the question of poles and zeros in DRPs and how the spectral properties of DRPs can be exploited for MIMO (iteration-domain) controller design

Future Work

- Future work can consider the question of poles and zeros in DRPs and how the spectral properties of DRPs can be exploited for MIMO (iteration-domain) controller design
- As in ILC, the problem of design for monotonic convergence comes up in DRP

Future Work

- Future work can consider the question of poles and zeros in DRPs and how the spectral properties of DRPs can be exploited for MIMO (iteration-domain) controller design
- As in ILC, the problem of design for monotonic convergence comes up in DRP
  - Consider the same plant used above, with the controller

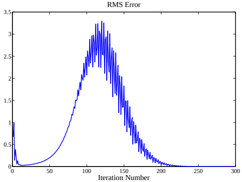
$$u_{k+1}(t) \quad = \quad 1.0806u_k(t) - u_{k-1}(t) + e_k(t) - e_{k-1}(t) + 0.667e_{k-2}(t)$$

Future Work

- Future work can consider the question of poles and zeros in DRPs and how the spectral properties of DRPs can be exploited for MIMO (iteration-domain) controller design
- As in ILC, the problem of design for monotonic convergence comes up in DRP
  - Consider the same plant used above, with the controller

$$u_{k+1}(t) = 1.0806u_k(t) - u_{k-1}(t) + e_k(t) - e_{k-1}(t) + 0.667e_{k-2}(t)$$

  - In this case the convergence exhibits the "wave phenomena" well-known in ILC



Further reading on DRP control

- "Internal Model Principle for DRPs," Kevin L. Moore and El-Sharif Omer, *Proc. 2009 IEEE Conference on Decision and Control*, pp. 446-451, Shanghai, China, December 2009
- "Iteration-Domain Closed-Loop Frequency Response Shaping for DRPs," Kevin L. Moore and Fadel Lashhab, *Proc. of 2010 American Control Conference*, Baltimore, MD, June 2010
- "Model Reference Adaptive Control of DRPs in the Iteration Domain," Kevin L. Moore and El-Sharif Omer, *Proc. of 2010 IEEE Int. Symp. on Intelligent Control*, Tokyo, Japan, September 2010
- "Norm-Optimal Control of Time-Varying DRPs," El-Sharif A. Omer and Kevin L. Moore, *Proc. of 2011 IEEE Int. Symp. on Intelligent Control*, Denver, Colorado, Sept 2011
- "Norm-Optimal Control of Time-Varying DRPs with Iteration-Varying Reference Inputs," El-Sharif A. Omer and Kevin L. Moore, *Proc. of 7th Int. Workshop on Multidimensional (nD) Systems*, Portiers, France, Sept 2011
- "Adaptive Reduced-Order Control of DRPs with Iteration-Varying Reference Signals," El-Sharif Omer, Kevin L. Moore, *Proc. of 2012 IEEE Int. Symp. on Intelligent Control*, Dubrovnik Palace Hotel, Dubrovnik, Croatia, October 3-5, 2012

Many Problems to Consider

Y_d	D	N	C	H	
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H_p	Classical Arimoto algorithm
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	$H_p(w)$	Owens' multipass problem
$Y_d(z)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	$N(w)$	$C(w)$	$H_p(w)+\Delta(w)$	**Most general problem**
$Y_d(z)$	$D(z)$	0	$C(w)$	$H(w)+\Delta(w)$	Frequency-domain uncertainty
$Y_d(z)$	0	$w(t), v(t)$	$\Gamma(w-1)^{-1}$	H_p	Least quadratic ILC
$Y_d(z)$	0	$w(t), v(t)$	$C(w)$	H_p	Stochastic ILC (general)
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H^I	Interval ILC
$Y_d(z)$	$D(z)$	$w(t), v(t)$	$C(w)$	$H(z)+\Delta H(z)$	Time-domain H_∞ problem
$Y_d(z)$	$D(w)$	$w(k,t), v(k,t)$	$C(w)$	$H(w)+\Delta H(w)$	Iteration-domain H_∞ ILC
$Y_d(z)$	0	$w(k,t), v(k,t)$	$\Gamma(k)$	$H_p+\Delta H(k)$	Iteration-varying uncertainty and control
$Y_d(z)$	0	$\widetilde{H}$	Γ	H_p	Intermittent measurement problem
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Concluding Comments

- DRP and ILC (a special case) are interesting repetitive (multi-pass) processes

Concluding Comments

- DRP and ILC (a special case) are interesting repetitive (multi-pass) processes
- We have shown how
  - An operator transform in the iteration domain leads to an algebraic framework for both ILC and DRP control processes
  - Ideas from MIMO control theory can be applied in this framework, such as the internal model principle and MRAC and more

Concluding Comments

- DRP and ILC (a special case) are interesting repetitive (multi-pass) processes
- We have shown how
  - An operator transform in the iteration domain leads to an algebraic framework for both ILC and DRP control processes
  - Ideas from MIMO control theory can be applied in this framework, such as the internal model principle and MRAC and more
- Thanks for listening!